TEACHING WALLAH

Teaching Exam

Subject – Computer Science

Computational Thinking and Programming 2

Lecture No -01

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Topics to be Covered

Topic

$\boxed{\text{Functions}}$

1hr

File Handling

$\boxed{\text{Python}}$

Function

(piece of code) which are reusable

b=
a=

(c= a* b)

prod(5, 4)

prod(d, e)

```
def prod(a, b):
    '''
    c=a*b
    return c
```

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Components.

def :

function_name :

parameters.

docstring

statements

return.

```
def value_add( a , b ):
    """ - - - - - - - - - - """
    c = a+b
    return c
```

parameters

value_add(2+3j , "Fine")

Arguments Arguments

value_add(- -)

0

default arguments

def value_add(a=0, b=0)

(a=0, b) ×

(b , a=0)

Functions.

Inbuilt functions.

len( )

print( )

type( )

User Defined Function.

Variables

Local variables →

Global variables →

```
x = 10        # Global

def mul(y):
    c = x * y     # c, y → Local variable
    return (print(c))

mul(5)
x
```

lambda function

$$\underline{\text{lambda}}\ \underbrace{\text{argument}}\ :\ \text{expression}$$

```
def square(x):
    return x*x
```

```
square = lambda x : x*x

print(square(5))

25
```

WAP to print the length of list

```
list1 = ["fine", 2+3j, 5, 2.5, "Hello"]

def len_list( list ):
    print(len(list))


len_list(list1)
    5
```

WAP to make a function which calculate factorial of n:

n=0, n==1

```
def fact(n):
    if n<0
        print("Error")
    elif (n==0 or n==1)
        return 1
    else
        return n*fact(n-1)
```

6

fact(-5)

Error

fact(0)

1

fact(3)

3 * f(2)

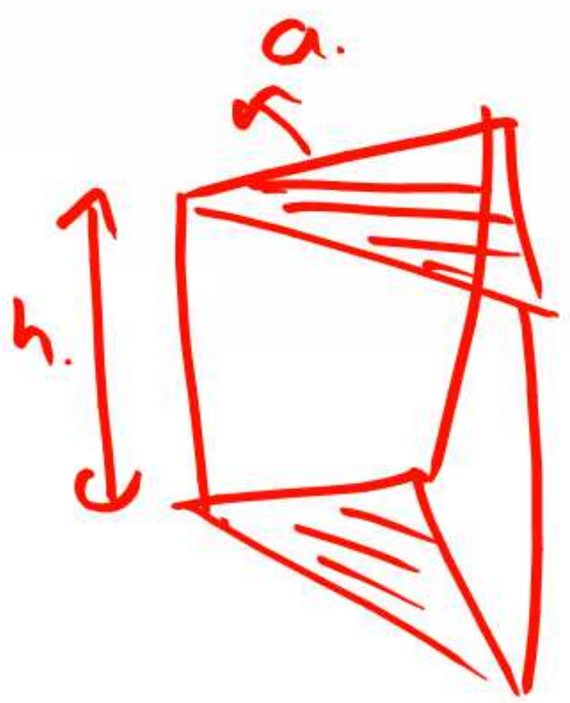
2 * f(1)

1

WAP to make a function which calculate volume of Equilateral Δ based prism.

area of eq $\Delta = \frac{\sqrt{3}}{4} a^2$

$= \underline{h}$



vol.= area of base × h.

2 mins Summary

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Topic

$\boxed{\text{File handling}}$

THANK YOU